



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 01

Target : JEE (Advanced) 2023

DPP Syllabus :

Geometrical Optics, Electrostatics, Gravitation

DPP No. : 01

Max. Marks : 156

Total Time : 146 Mins.

PART - I

One or more than one may be correct option ('-1' Negative marking) Q.1 to 13; (3 Marks; 2 Min.) [39, 26]

Subjective questions ('-1' Negative marking) Q.14 to Q.27; (4 Marks; 5 Min.) [56, 70]

Comprehension ('-1' Negative Marking) Q.28 to Q.32 (3 marks 2 min.) [15, 10]

Match the Following (no negative marking) Q.33 (8 marks 10 min.) [08, 10]

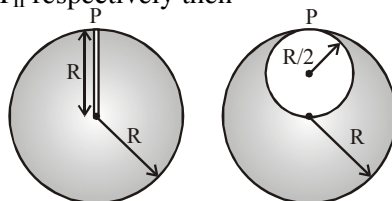
PART - II

One or more than one may be correct option ('-1' Negative marking) Q.34 to 41; (3 Marks; 2 Min.) [24, 16]

Subjective questions ('-1' Negative marking) Q.42 to Q.43; (4 Marks; 5 Min.) [08, 10]

Comprehension ('-1' Negative Marking) Q.44 to Q.45 (3 marks 2 min.) [06, 04]

1. Consider two solid spherical asteroid of uniform density of mass M and radius R . In one asteroid a tunnel of very small size of depth R is bored to the centre and in other asteroid a spherical cavity of radius $R/2$ is made as shown in the figure. Now, identical particles of mass m dropped into the cavities of both asteroids from the top most point P . If force experienced by particle is F_I and F_{II} respectively in cavities of asteroids I and II, when they are x distance away from the centre of asteroids. If the time taken by particles to reach the centre of asteroids is T_I and T_{II} respectively then

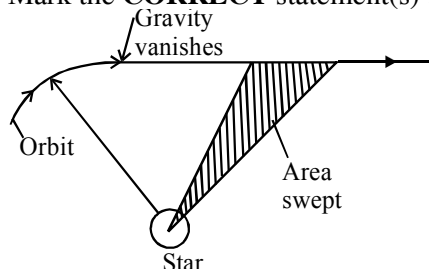


Asteroid - I

Asteroid - II

(A) the ratio of F_I/F_{II} is equal to $2x/R$ (B) the ratio of F_I/F_{II} is equal to x/R (C) the ratio of T_I/T_{II} is equal to $\pi/4$ (D) the ratio of T_I/T_{II} is equal to $\pi/2$

2. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :



(A) Newton's first law is obeyed on planet after gravity vanishes

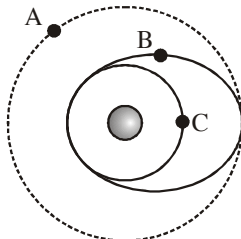
(B) Kepler's law of areas is obeyed only till the planet is in gravity of star

(C) Kepler's law of areas is obeyed even after gravity vanishes

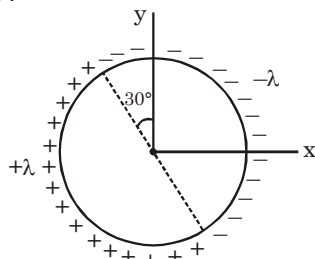
(D) Angular momentum of planet about centre of star is conserved through out its motion

3. A small satellite of mass 'm' is revolving around earth in a circular orbit of radius r_0 with speed v_0 . At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}(3/5)$ by turning its velocity vector in the same plane of motion, such that speed remains same. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is :-
 (A) 3 (B) 9 (C) 1/3 (D) 1/9

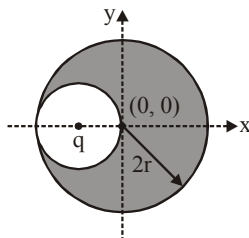
4. Three equal mass satellites A, B, and C are in coplanar orbits around a planet as shown in the figure. The magnitudes of the angular momenta of the satellites as measured about the planet are L_A , L_B , and L_C . Which of the following statements is correct?



- (A) $L_A > L_B > L_C$ (B) $L_C > L_B > L_A$ (C) $L_B > L_C > L_A$ (D) $L_B > L_A > L_C$
5. Consider a non-conducting ring of radius 'a' in x-y plane. The two semi-circular portions have linear charge densities $+\lambda$ and $-\lambda$ as shown. Then :

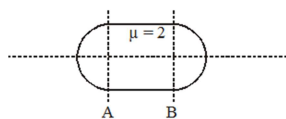


- (A) Dipole moment of the ring is $-(2\sqrt{3} \hat{i} + 2 \hat{j}) \lambda a^2$
 (B) Dipole moment of the ring is $-(2 \hat{i} + 2\sqrt{3} \hat{j}) \lambda a^2$
 (C) Electric field at the center of the ring is $\frac{\lambda}{2\pi\epsilon_0 a} (\sqrt{3} \hat{i} + \hat{j})$
 (D) Electric field at the center of the ring is $\frac{\lambda}{2\pi\epsilon_0 a} (\hat{i} + \sqrt{3} \hat{j})$
6. A spherical cavity of radius r is made in a neutral conducting sphere of radius $2r$. A charge q is kept at the centre of cavity as shown in the figure. Find the magnitude of the total electric field at $(4r, 0, 0)$.

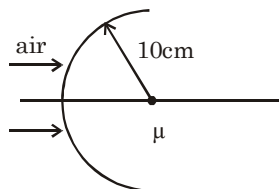


- (A) $\frac{Kq}{16r^2}$ (B) $\frac{4Kq}{r^2}$ (C) $\frac{Kq}{25r^2}$ (D) $\frac{4Kq}{9r^2}$

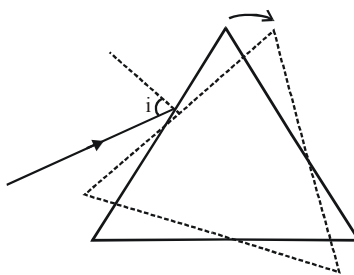
7. There is a uniform capsule shaped (hemispherical on sides and cylindrical in between) transparent glass, with $\mu = 2$, and radius of curvature 10 cm (for spherical portion) kept in air :-



- (A) For $AB = 20$ cm, a narrow parallel beam entering from left side will come out from other side as a parallel beam.
- (B) For $AB = 20$ cm, and surrounding air being replaced by a medium, $\mu = 3/2$, on both sides, light will come out as a diverging beam.
- (C) For $AB = 20$ cm, and surrounding air being replaced by a medium, $\mu = 3/2$, on both sides, light will come out as a parallel beam.
- (D) For surrounding $\mu = 3/2$, if emergent beam has to be parallel then AB has to be bigger than 20 cm.
8. A spherical surface separates air & medium for which $\mu_v = 1.615$ for violet and $\mu_R = 1.600$ for red color. A paraxial beam parallel of optic axis is incident on the surface as shown. The distance between point of convergence for violet and red color is Δf .

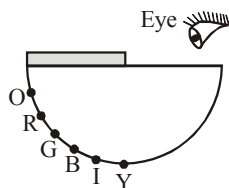


- (A) $\Delta f = 0.40$ cm
- (B) Point of convergence for red is closer to optical centre than that for violet.
- (C) Point of convergence for violet is closer to optical centre than that for red.
- (D) $\Delta f = 0.84$ cm.
9. An equilateral prism produces a deviation of 30° . When prism is rotated by an angle 30° , keeping the incident ray fixed as shown, angle of deviation remains same :-

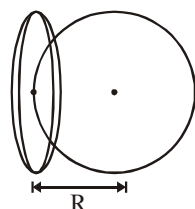


- (A) Angle of incidence before rotation is 45°
- (B) Angle of incidence before rotation is 30°
- (C) Initially prism is in a state producing minimum deviation
- (D) After rotation prism is in a state producing minimum deviation

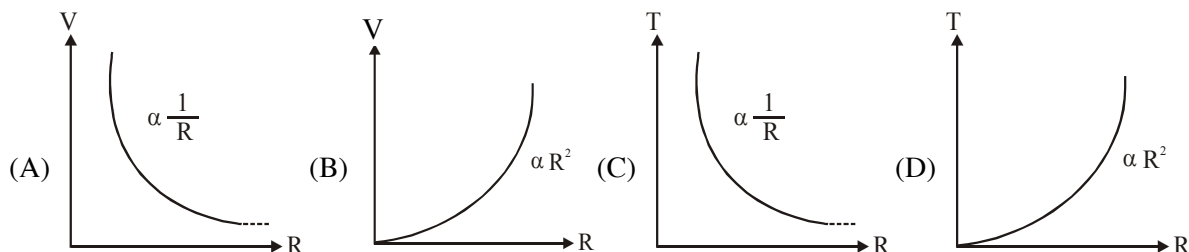
10. A person is looking at the flat surface of a transparent hemisphere. Half of the flat surface is coloured black and half of curved surface is coloured in six equal strips as shown in the figure. Then select correct alternative/s ($n_R = 1.412$, $n_O = 1.413$, $n_Y = 1.414$, $n_G = 1.415$, $n_B = 1.416$, $n_V = 1.417$) (where alphabates has usual meaning)



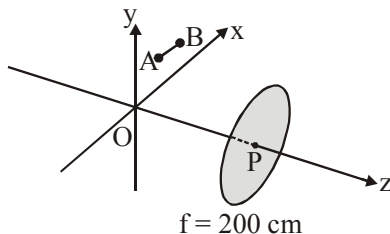
- (A) The person can see green orange and red
 (B) The person can see only yellow, indigo and blue
 (C) The ray coming from green, red and orange strips will be totally reflected
 (D) The ray coming from yellow, indigo and blue strips will be totally reflected
11. An uncharged conducting sphere of radius R is placed near a uniformly charged ring of radius R . Total charge on ring is Q . The centre of sphere lies on the axis of ring and distance of centre of sphere from centre of ring is R .



- (A) Potential at centre of ring is $\frac{KQ}{R}$
 (B) Potential at centre of ring is $\frac{KQ}{\sqrt{2}R}$
 (C) Potential at centre of ring due to conducting sphere is zero.
 (D) Potential at centre of conducting sphere is $\frac{KQ}{\sqrt{2}R}$
12. Imagine a light planet revolving around a very massive star in circular orbit of radius R in some other solar system, where gravitational force varies with r^{-3} (r is separation b/w the two particle) choose the correct curve(s) regarding motion of the planet. ($V \rightarrow$ speed of the planet, $T \rightarrow$ time period of planet)



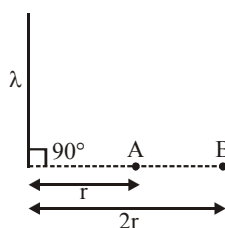
13. A converging lens of focal length 200 cm is placed parallel to x-y plane such that principal axis is along z-axis. The optical centre of the lens is at P (0, 0, 150 cm). Consider two points A(1 cm, 2 cm, 0) and B (2 cm, 2 cm, 0). The image of O, A and B is formed at O', A' and B' respectively, then :



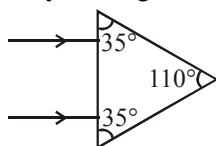
- (A) coordinates of A' = (4 cm, 8 cm, 800 cm) (B) coordinates of B' = (8 cm, 8 cm, -450 cm)
 (C) $\frac{\text{Area of } \triangle O' A' B'}{\text{Area of } \triangle OAB} = 16$ (D) the length A' B' = 4 cm

Integer Type

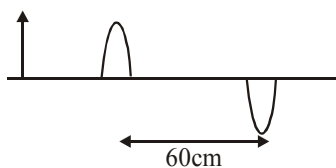
14. A man does 30 J of work on a ball to throw it vertically upto a height h on earth, if man does work x J on same ball to throw it vertically upto height $2h$ on another planet having its radius $\frac{1}{4}$ of radius of earth and planet has same density as that of earth. Find 'x' ? ($h \ll R_e$)
15. A binary star has a time period 3 years (time period of earth is one year) while distance between these two stars is 9 times distance between the earth and the sun. Mass of one star is equal to mass of the sun and mass of other star is $5n$ times mass of the sun then calculate n .
16. Four point charge $q, -q, 2Q$ and Q are placed in order at the corners A, B, C, D of a square. If the field at the mid point of CD is zero, then the value of q/Q is $\frac{5\sqrt{5}}{x}$. Find the value of x
17. In the figure shown points A and B are at distance r and $2r$ respectively from one end of uniformly charged infinitely long wire having linear charge density λ . If potential difference $V_A - V_B$ is $\frac{\lambda}{16\pi\epsilon_0} \ln(x)$, calculate x .



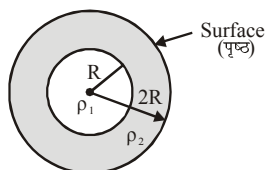
18. Two parallel beams of light pass through an isosceles prism of refractive index 1.4 as shown in the figure. Find the angle between the two emergent rays in degree. [$\sin 35^\circ = 0.57$, $\sin^{-1}(0.798) = 53^\circ$]



19. A converging lens (focal length f) is broken in two equal pieces and placed at 60 cm as shown along with the object. It is found that real images are formed at the same place and ratio of image heights is 9 : 1, if the value of f is $\left(x + \frac{1}{2}\right)$ cm then value of x is :



20. A planet is made of two materials of density ρ_1 and ρ_2 as shown in figure.

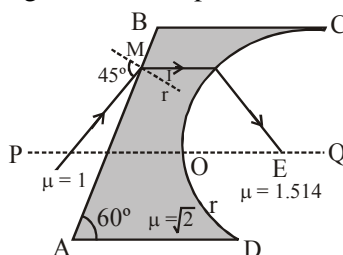


Acceleration due to gravity at surface of planet is same as at depth ' R '. The ratio $\frac{\rho_1}{\rho_2}$ is equal to $\frac{x}{6}$

Find the value of x .

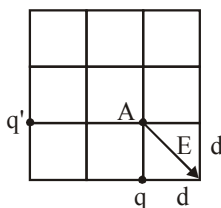
21. Equipotential surfaces differing in potential by 1 volt are drawn around a positive point charge. The distance of the nearest surface differing in potential by 1 V from the surface at 5 V is 10 cm. The distance of the other surface differing by 1 V from the 5 V surface is x cm, calculate x .

22. Figure shows an irregular block of material of refractive index $\sqrt{2}$. A ray of light strikes the face AB as shown in figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4 m, (with centre lying on the line PQ) and enter a medium of refractive index 1.514 to meet PQ at E. Find the distance OE in nearest interger in meters (point M is very near to line PQ)

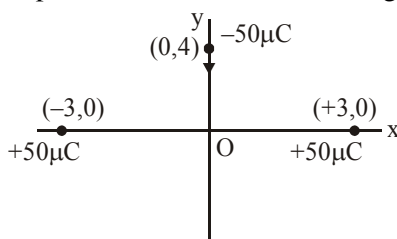


23. A planet revolves around the sun in elliptical orbit of semi-major axis 2×10^{12} m. The areal velocity of the planet when it is nearest to the sun is 4.4×10^{16} m²/s. The least distance between planet and the sun is 1.8×10^{12} m. Find the minimum speed of the planet in km/s.

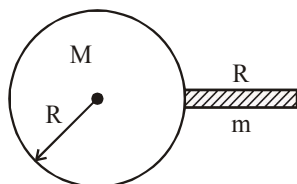
24. In given arrangement side of each small square is d . Two point charges q and q' are placed as shown in the figure. If the total electric field at A is as shown in the figure below and potential at point A is given by $\frac{-nq}{8\pi\epsilon_0 d}$ then find n .



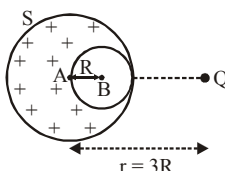
25. The figure shows two equal, positive charges, each of magnitude $50 \mu\text{C}$, fixed at points $(3, 0) \text{ m}$ and $(-3, 0) \text{ m}$ respectively. A charge $-50 \mu\text{C}$, moving along negative y -axis has a kinetic energy of 4 J at the instant it crosses point $(0, 4) \text{ m}$. If the y co-ordinate of this charge where direction of its motion reverses for the first time after crossing this point is $x\sqrt{2} \text{ m}$ then, x is (neglect gravity).



26. A uniform thin rod of mass m and length R is placed normally on surface of earth as shown. The mass of earth is M and its radius is R . If the ratio of magnitude of gravitational force exerted by earth on the rod and the magnitude of gravitational potential energy of the rod and earth system is $\frac{1}{\mu R \ln 2}$ then ' μ ' is

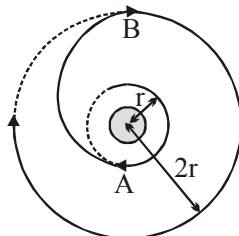


27. A point charge Q is kept at distance $r = 3R$ from the centre (point A) of a uniformly charged sphere having volume charge density ρ and radius $2R$. The sphere has a spherical cavity of radius R having centre at point B such that $AB = R$. If the interaction energy between the point charge and the sphere having cavity is given as $\frac{13Q\rho R^2}{N\epsilon_0}$. Find N ? (Line joining points A and B passes through point charge Q)



Paragraph for Question Nos. 28 to 29

The Hohmann Transfer Orbit maneuver is used by spaceships to transfer from one circular orbit around a planet to another, by temporarily entering an intermediate elliptical orbit. The spaceship starts on a small circular orbit of radius r around a planet of mass M . At point A in the diagram below, it fires its propellers to put itself onto an intermediate elliptical orbit. At point B, it fires its propellers again, and ends on the larger circular orbit of radius $2r$. You may assume that the propellers are so powerful that the changes in velocity occur instantaneously at each point. Between point A and B space ship experience only gravitational force from planet.



28. Find ΔV_A (the amount by which the velocity needs to be changed at point A) needed to achieve this travel path.

(A) $\sqrt{\frac{GM}{r}} \left(\sqrt{\frac{2}{3}} - 1 \right)$ (B) $\sqrt{\frac{GM}{r}} \left(\sqrt{\frac{4}{3}} - 1 \right)$ (C) $\sqrt{\frac{GM}{r}} \left(\sqrt{\frac{4}{3}} + 1 \right)$ (D) $\sqrt{\frac{2GM}{r}} \left(\sqrt{\frac{4}{3}} - 1 \right)$

29. Time period of a satellite in inner orbit is 2 days then what is find taken by space ship to go from A to B.

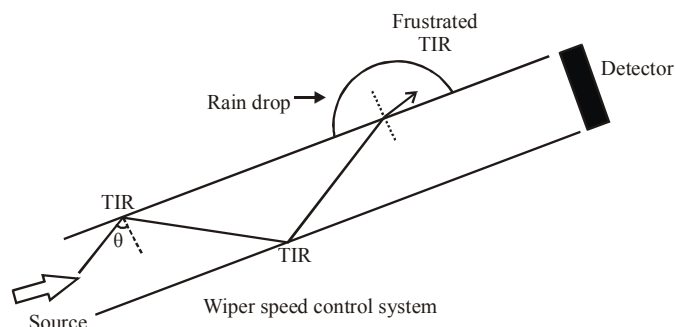
(A) $(2)^{3/2}$ days (B) $\left(\frac{4}{3}\right)^{3/2}$ days (C) $\left(\frac{3}{2}\right)^{3/2}$ days (D) $\left(\frac{2}{3}\right)^{3/2}$ days

Paragraph for Question No. 30 to 32

Frustrated TIR (F-TIR) : In optics, when light rays traveling in a denser medium strike at medium boundary at an angle greater than critical angle, TIR occurs and a surface wave which is called **Evanescence wave** forms in rarer medium. An evanescent wave is a near-field standing wave with an intensity that exhibits exponential decay with distance (less than wavelength) from the boundary at which the wave is formed.

Imagine that a beam of light traveling within a block of glass is internally reflected at a boundary. Presumably if you pressed another piece of glass against the first, the air-glass interface would be made to vanish and the beam would then propagate onward undisturbed. Furthermore, this transition from total to no reflection occurs gradually as the air film between them thinned out as explained above, when third medium with a higher refractive index (than the low-index second medium) is placed within less than several wavelengths distance from the interface between the first medium and the second medium, the evanescent wave will be different from the one under "ordinary conditions" and it will pass energy across the second into the third medium (evanescent wave coupling). This process is called "frustrated" total internal reflection (FTIR) and is very similar to quantum tunneling. An example of application of this principle is automatic **Wiper speed control** found in high end cars like Skoda, Audi, BMW etc.

Figure shows an example of an optical system designed to detect the amount of water present on the windshield of a car to adjust the wiper speed. As shown in this figure, we can use the windshield as a waveguide to guide the light from a source located at one end (bottom of the windshield) to a detector located in the opposite end. The light suffers total-internal reflection (TIR) at the glass-air interface. However, when rain drops are present, some of the light will suffer frustrated TIR escaping outside the waveguide. Since we know the power of the light source, a given drop in power can be correlated to the amount of water present and used to adjust the wiper speed.



30. Choose the **CORRECT** statement.

- (A) The energy carried by evanescent wave is zero in both TIR and FTIR.
- (B) The energy carried by evanescent wave is non-zero in both TIR and FTIR.
- (C) The energy carried by evanescent wave is zero in TIR and non-zero in FTIR.
- (D) The energy carried by evanescent wave is non-zero in TIR and zero in FTIR.

31. In the following systems which should not be based on FTIR?

- (A) Student card attendance system
- (B) Thumb attendance system
- (C) Finger print scanner
- (D) Multi touch screen

32. For the wiper speed control system to work, the angle of incidence on the glass air interface is θ , then

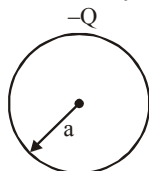
- (A) $\theta < \sin^{-1}\left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}}\right)$
- (B) $\theta > \sin^{-1}\left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}}\right)$
- (C) $\sin^{-1}\left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}}\right) < \theta < \sin^{-1}\left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}}\right)$
- (D) $\theta > \sin^{-1}\left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}}\right)$

33. In each situation of list-I, some charge distributions are given with all details explained. In list-II The electrostatic potential energy and its nature is given situation in list-II. Match the proper entries from list-II to list-I using the codes given below the lists.

List-I

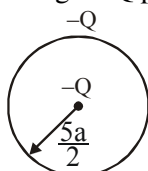
- (P) A thin shell of radius 'a' and having a charge

$-Q$ uniformly distributed over its surface as shown.



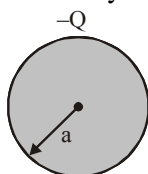
- (Q) A thin shell of radius $\frac{5a}{2}$ and having a charge $-Q$

uniformly distributed over its surface and a point charge $-Q$ placed at its centre as shown.

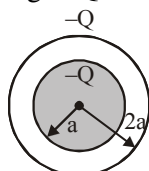


- (R) A solid sphere of radius 'a' and having a charge $-Q$

uniformly distributed throughout its volume as shown.



- (S) A solid sphere of radius 'a' and having a charge $-Q$ uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius $2a$ and carrying charge $-Q$ as shown

**List-II**

(1) $\frac{1}{8\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

(2) $\frac{3}{20\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

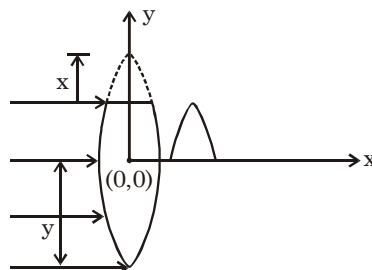
(3) $\frac{27}{80\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

(4) Positive in sign

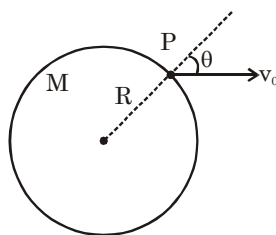
	(P)	(Q)	(R)	(S)
(A)	1	2	4	3
(B)	4	1	2	3
(C)	1	2	3	4
(D)	2	4	3	1

PART -II

34. The dotted part of the converging lens is cut and kept on the x axis as shown in the diagram. If parallel paraxial rays are falling on this system then the coordinate of image formed after refraction from both the lenses is $(30, -1)$. If $x = 2.5$ then (all the distance are in cm) (Assume lens have no spherical aberration) :-

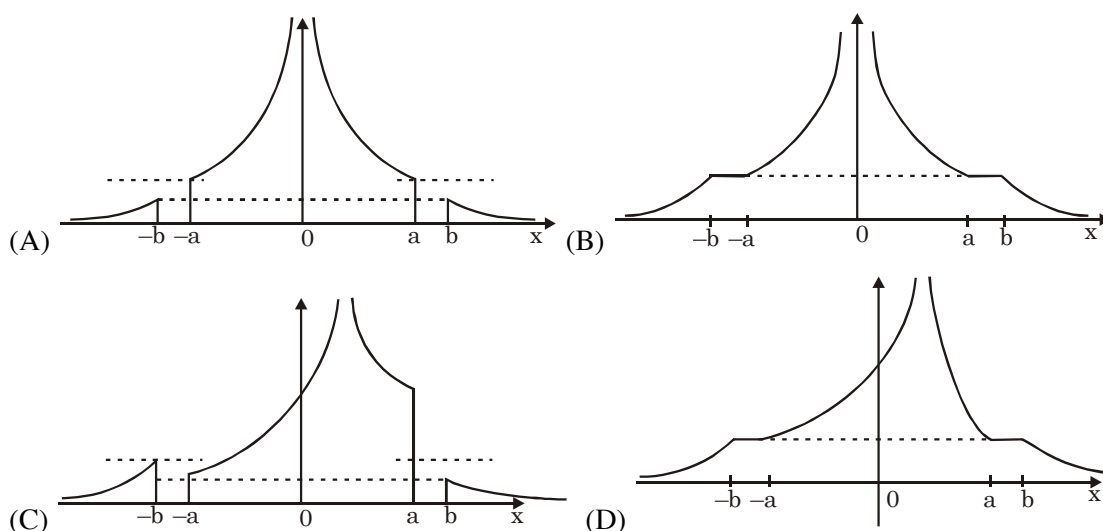
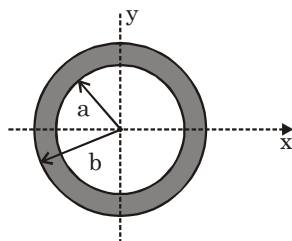


- (A) $y = 2$ cm (B) $y = 6$ cm (C) $y = 4$ cm (D) $y = 4.5$ cm
35. Consider a planet of mass M and radius R as shown in the figure. A particle is projected from the surface of the planet at an angle θ from radial direction and with velocity equal to the orbital velocity of circular orbit at that point then choose the **CORRECT** statement(s) :

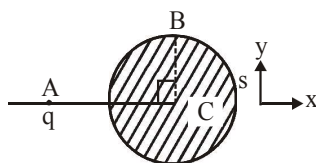


- (A) Maximum distance of particle from centre of the planet is $(R + R \cos\theta)$.
- (B) Path of the particle is parabola.
- (C) Time taken by the particle to return to the surface of planet is $\sqrt{\frac{R^3}{GM}}$.
- (D) Time taken by the particle to reach the surface of planet is more than $\pi\sqrt{\frac{R^3}{GM}}$.

36. A positive point charge q is located inside a neutral hollow spherical conducting shell. The shell has inner radius a and outer radius b ; $b - a$ is not negligible. The shell is centred on the origin. Which of the following is correct graph of electric field vs radial distance x or electric potential vs radial distance x . The point charge can be located anywhere inside shell on x -axis.

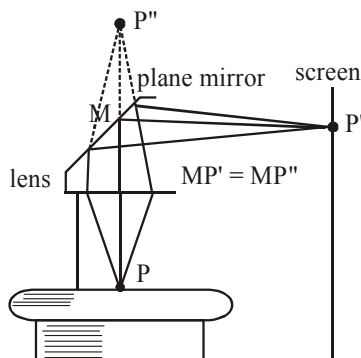


37. In the figure shown a point charge q is kept at a distance $4R/3$ from centre of a conducting solid sphere 'S' of radius R and having a charge Q . B is a point on the surface of conductor such that $BC \perp AC$. Also line AC is along X-axis. Choose the correct options.



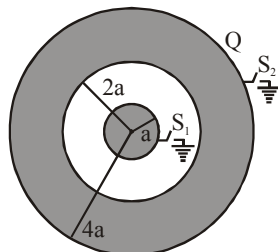
- (A) X-component of electric field at the point B only due to conductor is $\frac{9q}{125\pi\epsilon_0 R^2} \hat{i}$.
- (B) X-component of electric field at the point B only due to conductor is $\frac{-9q}{125\pi\epsilon_0 R^2} \hat{i}$.
- (C) Electric potential at the point B only due to conductor is $\frac{1}{4\pi\epsilon_0 R} \left[\frac{3q}{20} + Q \right]$.
- (D) Electric potential at the point C only due to conductor is $\frac{1}{4\pi\epsilon_0 R} \left[\frac{3q}{20} + Q \right]$.

38. The lens in an overhead projector forms an image P' of a point P on a screen as shown. If the screen is moved closer to the projector to keep the image on the screen in focus, then select possible correct option/s:

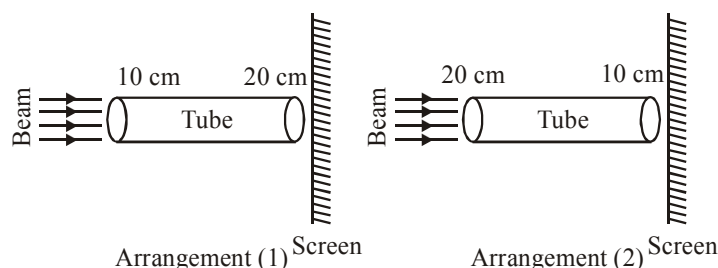


- (A) The lens is moved away from P keeping same principal axis.
 (B) The lens is moved towards from P keeping same principal axis.
 (C) The mirror is moved towards the lens keeping same orientation.
 (D) The mirror is moved away from the lens keeping same orientation.
39. Four point masses each of mass m are placed on vertices of a regular tetrahedron. Distance between any two masses is r .
- (A) Gravitation field at centre is zero.
 (B) Gravitation potential at centre is $\frac{-4Gm}{r}$.
 (C) Gravitation potential energy of system is $\frac{-6Gm^2}{r}$.
 (D) Gravitation force on one of the point mass is $\frac{\sqrt{6}Gm^2}{r^2}$.
40. Consider a spherical symmetric distribution of charge. The centre of symmetry is at O . Electric flux linked with a spherical Gaussian surface of radius ' r ' and centered at O is given by $\phi = \phi_0 r^4$. Choose the correct option(s).
- (A) Volume charge density at a distance ' r ' from O is given by $\frac{2\epsilon_0 \phi_0 r}{\pi}$.
 (B) Volume charge density at a distance ' r ' from O is given by $\frac{\epsilon_0 \phi_0 r}{\pi}$.
 (C) Electrostatic potential energy stored in the spherical region of radius r centred at O is given by $\frac{\epsilon_0 \phi_0^2 r^7}{28\pi}$.
 (D) Electrostatic potential energy stored in the spherical region of radius r centred at O is given by $\frac{\epsilon_0 \phi_0^2 r^7}{56\pi}$.

41. A thick hollow metallic sphere of inner and outer radii $2a$ and $4a$, respectively, encloses a solid concentric metallic sphere of radius ' a '. These spheres are connected to the earth through switches S_1 and S_2 as shown in the figure. First, switch S_1 is closed and after some time S_1 is opened. Then, S_2 is closed and after some time S_2 is opened. This completes one cycle. Initially, charge Q is given to the outer sphere with both S_1 and S_2 open. The charge on the inner sphere after n cycles of earthing the inner and outer spheres is

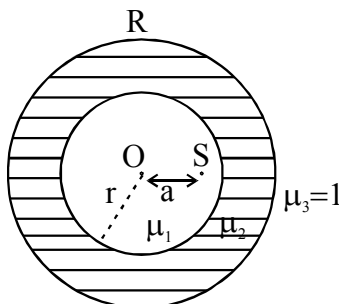


- (A) $-Q/3$ (B) $-Q/3^n$ (C) $Q/3^n$ (D) None
42. Two persons A and B wear glasses of optical powers (in air) $P_1 = +2$ D and $P_2 = +1$ D respectively. The glasses have refractive index 1.5. Now they jump into a swimming pool and look at each other. B appears to be present at distance 2 m (from A) to A. A appears to be present at distance 1 m (from B) to B. The refractive index of water in the swimming pool, in the form $\frac{X}{5}$ and find X.
43. A tube of length 30 cm has its inner lateral surface which is completely absorbing. Two lenses of focal lengths 10 cm and 20 cm are fixed at the ends of the tube. For same beam, intensities of spots on screen are I_1 and I_2 for arrangement (1) and arrangement (2) respectively. Find the value of $\frac{I_2}{I_1}$

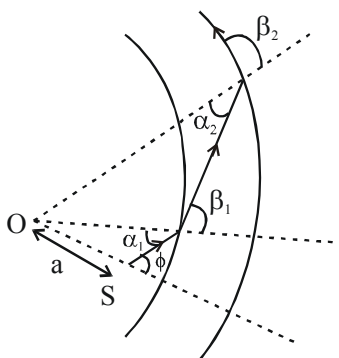


Paragraph For Questions 44 to 45

A glass spherical shell of centre O with an outer radius of $R = 7.5$ cm and an inner radius of $r = 6.5$ cm has a refractive index $\mu_2 = 1.5$. The inside of the shell is filled with a material, whose refractive index is $\mu_1 = \frac{25}{16}$. A source of light S is placed at a distance of $a = 6$ cm from the center.



44. Consider a light ray starting at S and getting double refracted into surrounding as shown with $\beta_2 = 90^\circ$. Dotted lines represent normals at the points of refraction. Which of the following statement(s) are **CORRECT**?



- (A) $\sin \beta_1 = \frac{R}{\mu_2 r}$ (B) $\sin \alpha_1 = \frac{R}{\mu_1 r}$ (C) $\frac{\sin \phi}{\sin \alpha_1} = \frac{r}{a}$ (D) $\sin \phi = \frac{R}{\mu_1 r}$
45. What percentage of the energy of the light source leaves from right half of the system into medium ? (Assuming no absorption of light inside the glass).
- (A) 20% (B) 40% (C) 37.5 % (D) 75 %

ANSWER KEY TO DPP # 01 (JEE-ADVANCED)

1.	(A, C)	2.	(A, C, D)	3.	(B)	4.	(A)	5.	(A, C)
6.	(A)	7.	(A,B,D)	8.	(A,C)	9.	(B)	10.	(B, C)
11.	(B, D)	12.	(A, D)	13.	(B, C, D)	14.	15	15.	16
16.	02	17.	16	18.	$X = 36$	19.	22	20.	14
21.	15	22.	6	23.	40	24.	02	25.	-6
26.	2	27.	18	28.	(B)	29.	(C)	30.	(C)
31.	(A)	32.	(C)	33.	(P) 1,4 ; (Q) 2,4 ; (R) 2,4 ; (S) 3,4				
34.	(D)	35.	(A, D)	36.	(A, B, D)	37.	(B, C)	38.	(A, D)
39.	(A,C,D)	40.	(B, D)	41.	(B)	42.	+6	43.	16
44.	(A,B,C)	45.	(A)						